



Oracle Database Administration

I

Chapter 2: Installing the Oracle Binaries

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Objectives

1. Describe your role as a database administrator (DBA) and explain typical tasks
2. Plan an Oracle Database installation
3. Use Optimal Flexible Architecture (OFA)
4. Install the Oracle software by using Oracle Universal Installer (OUI)

1. Describe your role as a database administrator (DBA) and explain typical tasks:

The approach for designing, implementing, and maintaining an Oracle database involves the following tasks:

1. Evaluating the database server hardware.
2. Installing the Oracle software.
3. Planning the database and security strategy.
4. Creating, migrating, and opening the database.
5. Backing up the database.
6. Enrolling system users and planning for their Oracle Network access.
7. Implementing the database design.
8. Recovering from database failure.
9. Monitoring database performance.

2. Plan managing database and tools:

Oracle database monitoring and management is not entirely different from the on premise database paradigm. In fact, many of the tools, such as SQL Developer, TOAD, and OEM 19c Cloud Control, allow you to connect to your cloud database and perform most of the tasks as before.

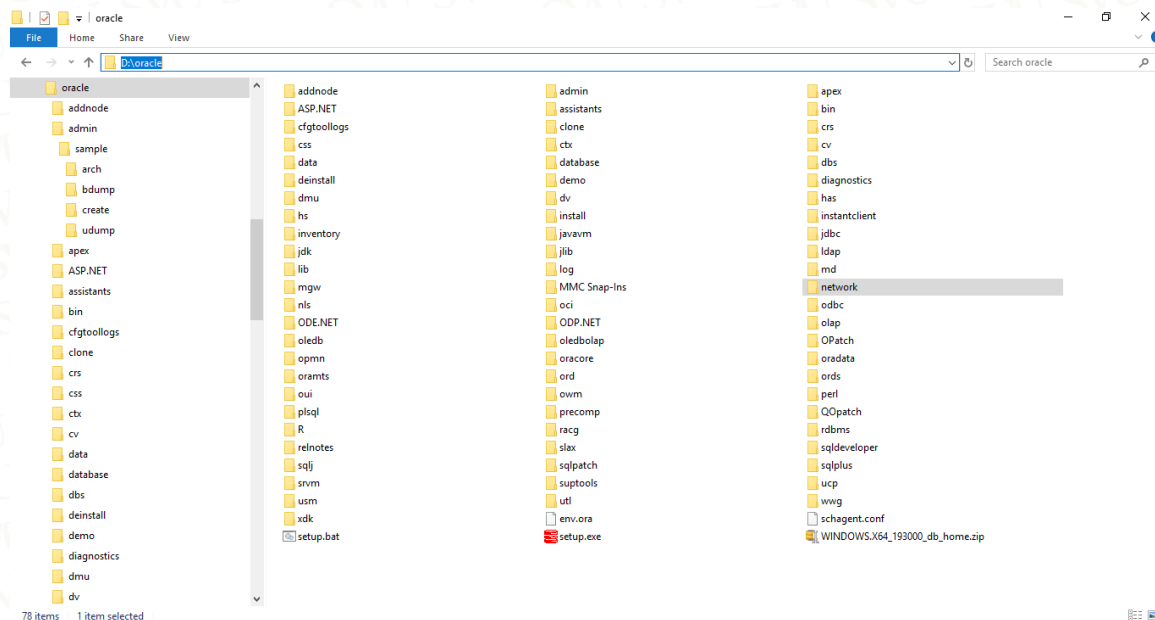
- Oracle Universal Installer
- Database Configuration Assistant
- Database Upgrade Assistant
- Oracle Net Manager
- Oracle Enterprise Express Manager
- SQL*Plus
- Recovery Manager
- Data Pump
- SQL*Loader
- Toad for Oracle

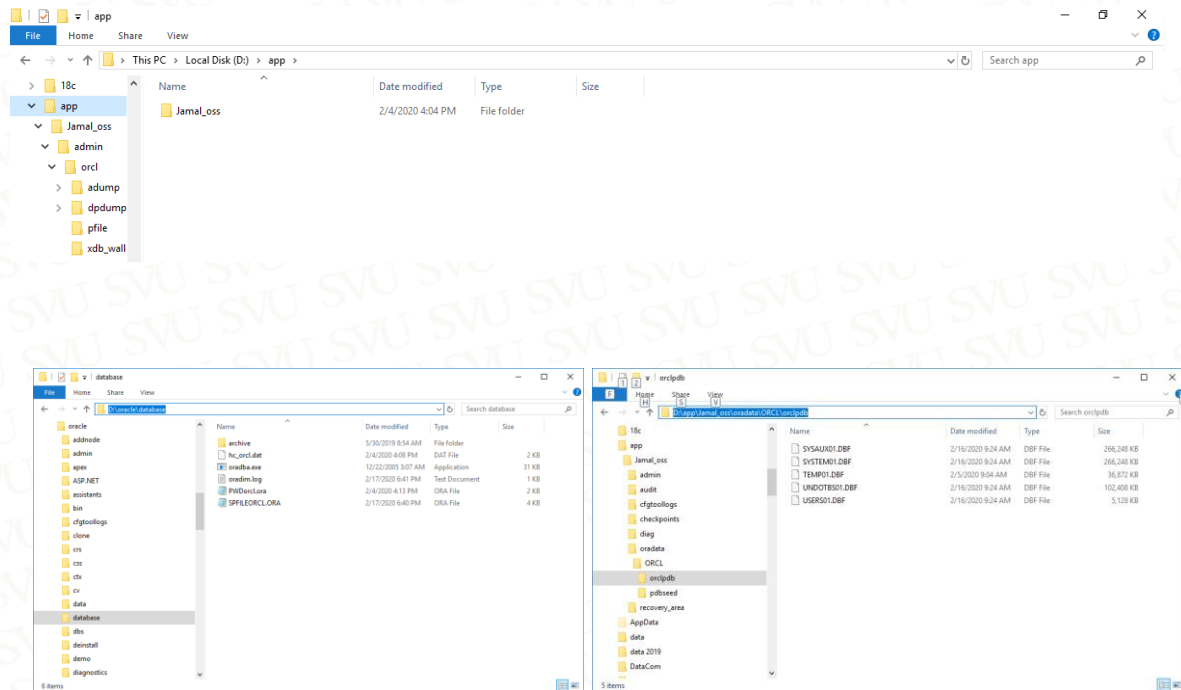
3. Use Optimal Flexible Architecture (OFA):

OFA is a file system directory structure that should make maintaining multiple versions of multiple Oracle products straightforward.

The OFA standard provides ways to understand where log files are available on a consistent basis. If standards are followed, security, migrations, and automations are going to be easier to implement because of consistency across the environments. The consistent locations of the log files allow for the files to be used by other tools as well as being secured. The ORACLE_BASE directory in 19c provides a way to separate the ORACLE_HOME directories for read-only directories and have the writable files in the ORACLE_BASE. Read-only ORACLE_HOME directories allow for implementing separation of installation and configuration, which is important for the cloud and securing the environment. This simplifies patching as one image can be used for a mass rollout and distribute a patch to many servers and reduces downtime for patching and updating of the Oracle software.

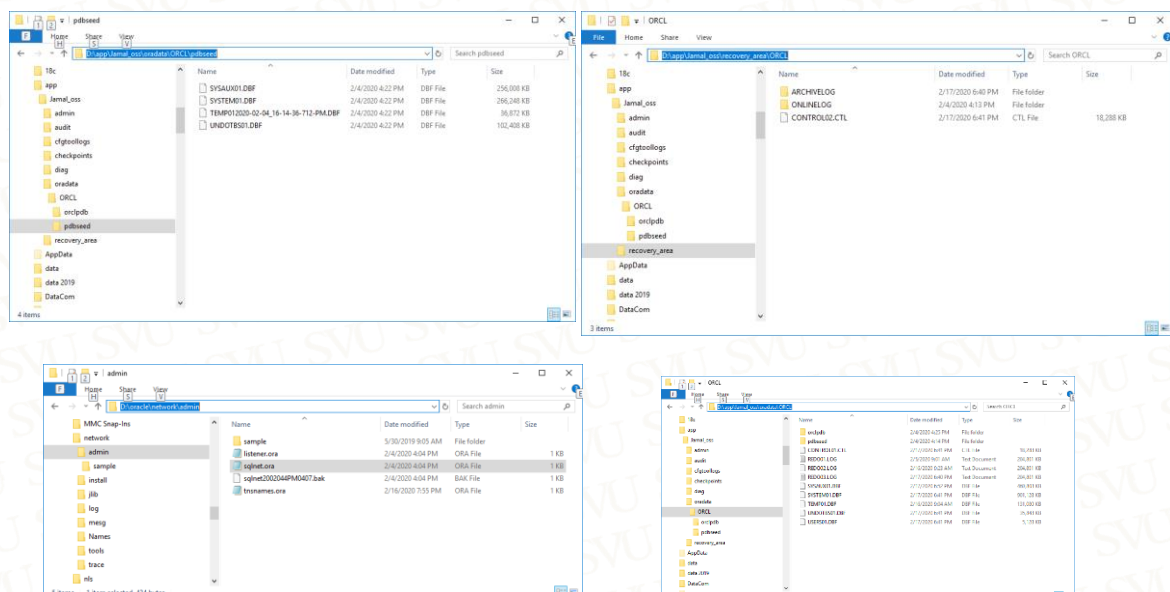
The following figures show the directory structures and file names used with the OFA standard. All the directories and files found in an Oracle environment appear in these figures:





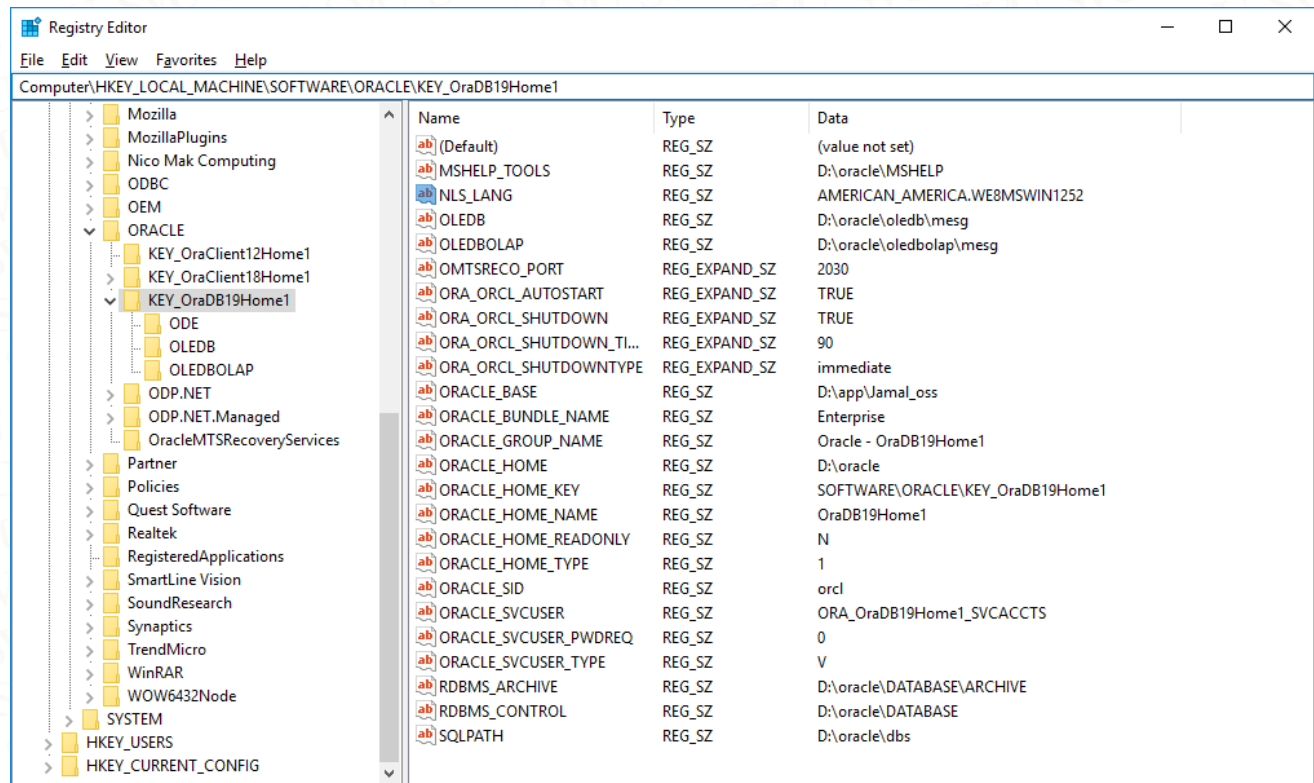
The OFA standard includes several directories that you should be familiar with:

- Oracle inventory directory
- Oracle base directory (ORACLE_BASE)
- Oracle home directory (ORACLE_HOME)



- Oracle network files directory (TNS_ADMIN)
- Automatic Diagnostic Repository (ADR_HOME)

See the below figure from windows registry related to Oracle 19c:



4. Install Oracle Database 19c on Windows:

We will learn how to install Oracle Database 19c on Windows 10 Enterprise Step By Step. Before we start installing, we need to know the minimum requirements for Oracle Database 19c:

Windows x64 Minimum Hardware Requirements

- System Architecture: Processor: AMD64 and Intel EM64T at least
- Physical memory (RAM): 8 GB minimum to run the core processes normally without any impact on system performance, knowing that 2 GB is sufficient to run Oracle alone
- Virtual memory (swap):
 - If physical memory is between 2 GB and 16 GB, then set virtual memory to 1 times the size of the RAM
 - If physical memory is more than 16 GB, then set virtual memory to 16 GB
- Disk space:
 - Typical Install Type total: 10 GB
 - Advanced Install Types total: 10 GB
 - *Oracle recommends that you allocate approximately 100 GB to allow additional space for applying any future patches on top of the existing Oracle home.*
- Video adapter: 256 colors
- Screen Resolution: At least 1024 x 768 display resolution, which Oracle Universal Installer requires
- It's important to know that Oracle Database 19c is only available for 64 bit Windows systems (Workstation or Server). Therefore, we can't install Oracle Database 19c in our PC if it's running a 32 bit Windows OS.
- Oracle Database 19c is only available for Windows 10 Professional, Enterprise and Education Editions. This means we can't install Oracle Database 19c if we are using Windows 10 Home Edition
- To install Oracle Database 19c we have to download the software, which we can do by visiting the OTN page:

<https://www.oracle.com/database/technologies/oracle-database-software-downloads.html#19c>

- Before beginning the installation of Oracle Database 18c, we need to verify that we are connected to our Windows as an administration user. If we are not, we must log in to our Windows as an administrative user. Not doing so will cause an installation error and crash the entire process.
- Mount point paths for the software binaries: Oracle recommends that you create an **Optimal Flexible Architecture configuration**

Verifying Hardware Requirements

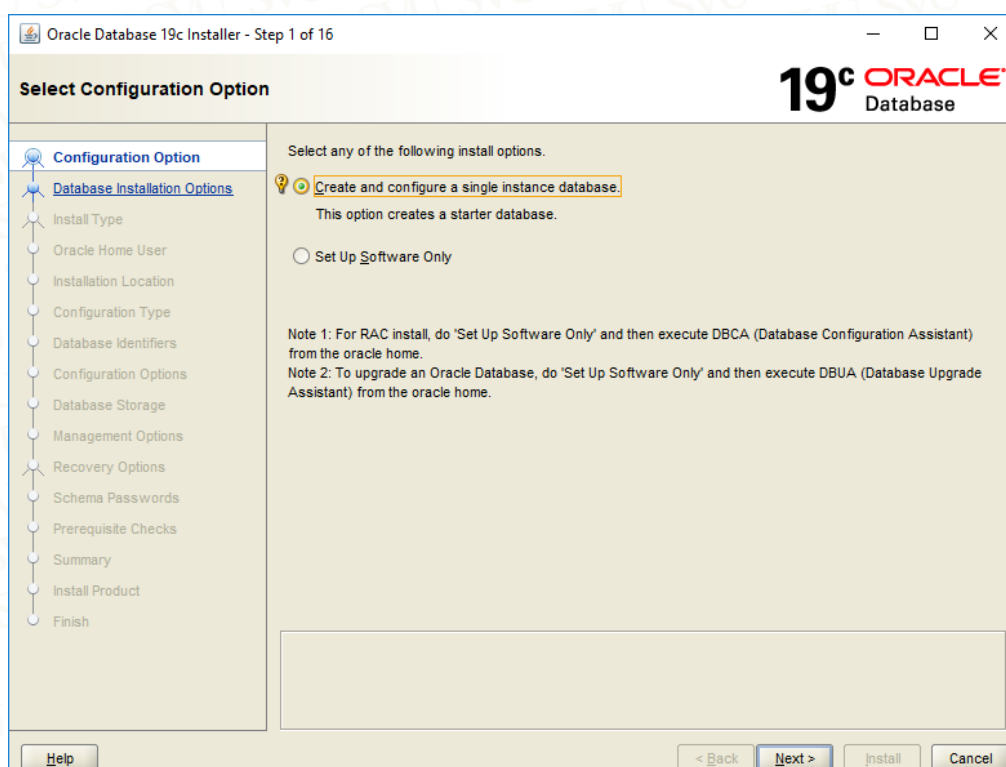
1. Determine the physical RAM size: If the size of the physical RAM installed in the system is less than the required size, then you must install more memory before continuing.
2. Determine the size of the configured virtual memory (also known as paging file size).
3. Determine the amount of free disk space on the system.
4. Determine the amount of disk space available in the temp directory. This is equivalent to the total amount of free disk space, minus what is required for the Oracle software to be installed.

Oracle Universal Installer (OUI)

To begin the installation of the Oracle Database 19c, we must first go to the **Oracle19c** folder; for example, in my case, D:\Oracle. and then locate the setup.exe file. Run the **setup.exe** file and choose run as administrator.



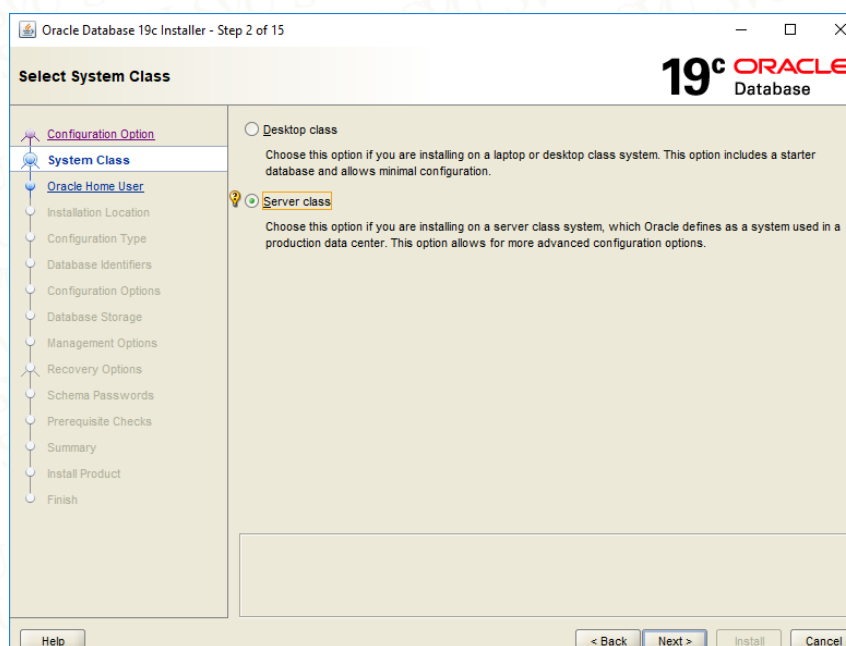
Here we are presented with two installation options “Configuration Option”:



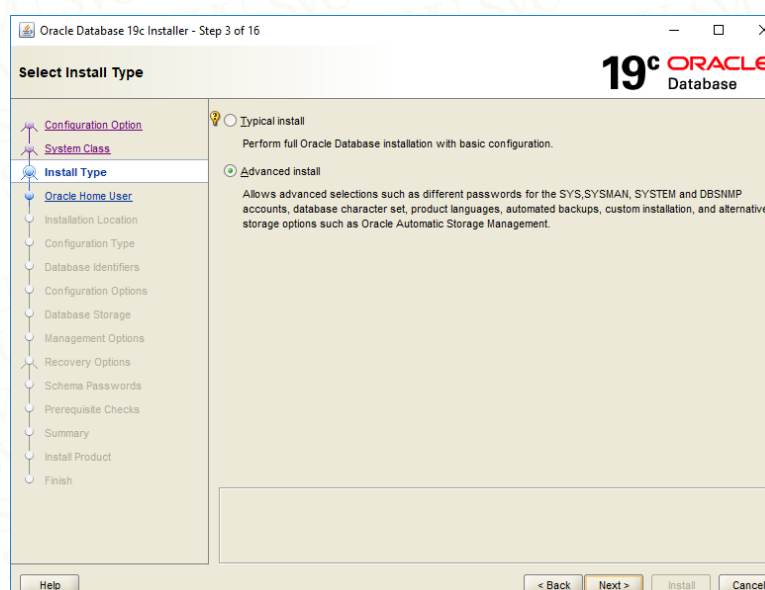
The first is to “Create and configure a single instance database”. Choosing this option will not only configure an Oracle Database Server on our machine but also create a starter database for us. We can choose this database for learning and practice purposes. This is our choice now

The second option is “Set Up Software Only”. As the name suggests, it will only configure the Oracle Server software on our machine. No database will be created. We need to create the database manually using the DBCA utility. We choose this option when we are configuring a RAC or planning to perform an upgrade.

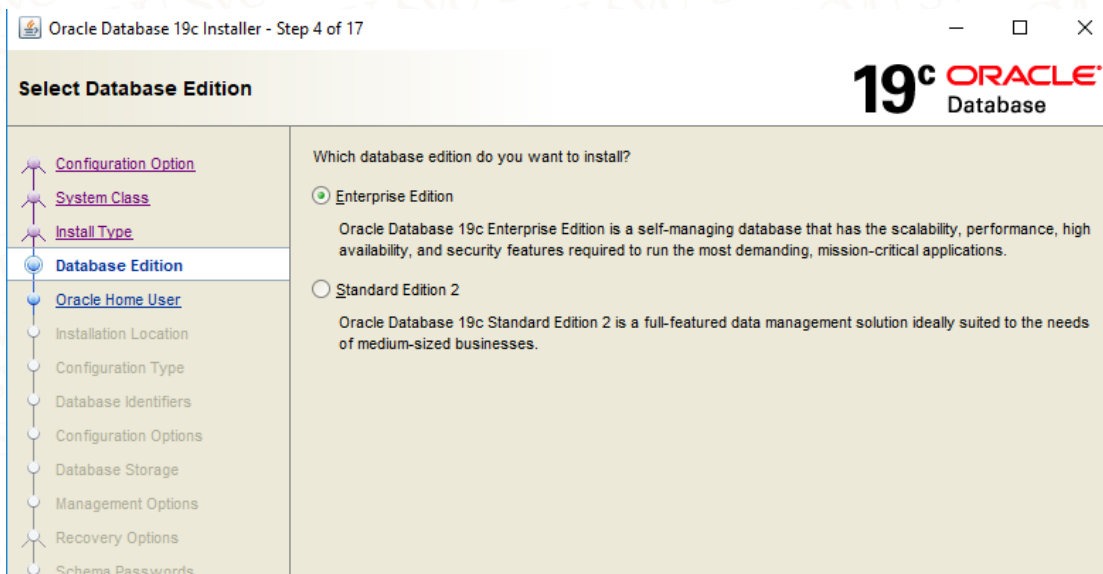
Then we presented “System Class” options: **Desktop class and Server Class as below:**



Install Type step:



Database Edition Type step:



Oracle Database 19c Installer - Step 4 of 17

Select Database Edition

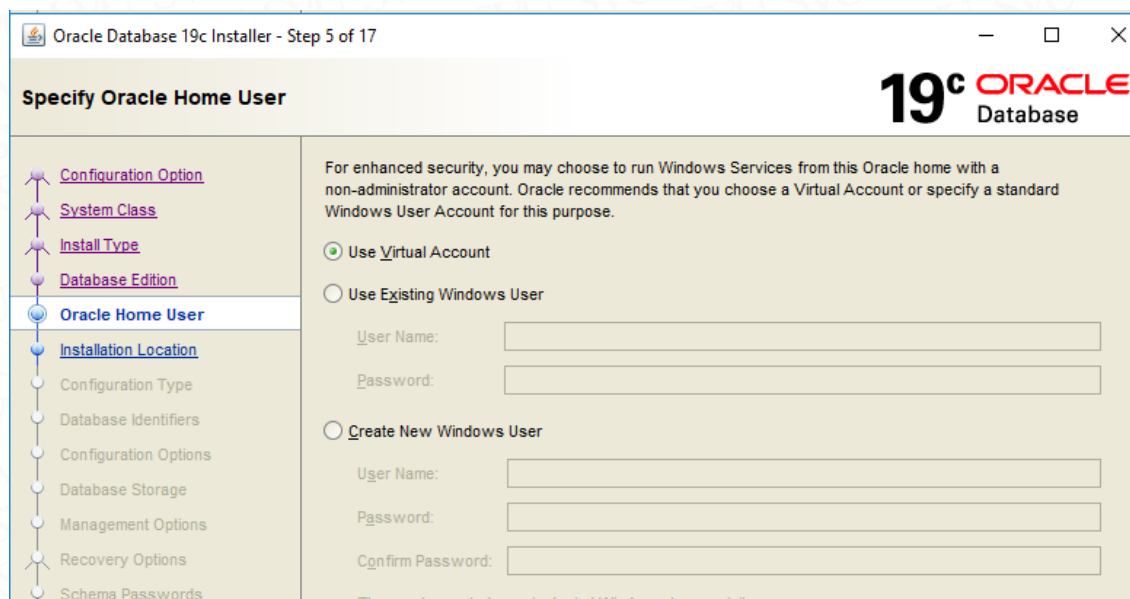
Which database edition do you want to install?

☒ **Enterprise Edition**
 Oracle Database 19c Enterprise Edition is a self-managing database that has the scalability, performance, high availability, and security features required to run the most demanding, mission-critical applications.

☐ **Standard Edition 2**
 Oracle Database 19c Standard Edition 2 is a full-featured data management solution ideally suited to the needs of medium-sized businesses.

Configuration Option
 System Class
 Install Type
Database Edition
 Oracle Home User
 Installation Location
 Configuration Type
 Database Identifiers
 Configuration Options
 Database Storage
 Management Options
 Recovery Options
 Schema Passwords

Oracle Home User:



Oracle Database 19c Installer - Step 5 of 17

Specify Oracle Home User

For enhanced security, you may choose to run Windows Services from this Oracle home with a non-administrator account. Oracle recommends that you choose a Virtual Account or specify a standard Windows User Account for this purpose.

☒ **Use Virtual Account**
☐ **Use Existing Windows User**
☐ **Create New Windows User**

User Name:
 Password:
 Confirm Password:

Configuration Option
 System Class
 Install Type
 Database Edition
Oracle Home User
 Installation Location
 Configuration Type
 Database Identifiers
 Configuration Options
 Database Storage
 Management Options
 Recovery Options
 Schema Passwords

Installation Location:

Oracle Database 19c Installer - Step 6 of 17

Specify Installation Location

19c ORACLE Database

Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.

Oracle base?

This software directory is the Oracle Database home directory.

Software location: D:\oracle

Configuration Type:

Oracle Database 19c Installer - Step 7 of 17

Select Configuration Type

19c ORACLE Database

Select the type of database that you want to create.

☒ General Purpose / Transaction Processing
A starter database designed for general purpose use, or for transaction-heavy applications.

☐ Data Warehousing
A starter database optimized for data warehousing applications.

Database Identifiers:

Oracle Database 19c Installer - Step 8 of 17

Specify Database Identifiers

19c ORACLE Database

Provide the identifier information required to access the database uniquely. An Oracle database is uniquely identified by a Global database name, typically of the form "name.domain". A database is referenced by at least one Oracle instance which is uniquely identified from any other instance on this computer by an Oracle system identifier (SID).

Global database name:

Oracle system identifier (SID):

☒ Create as Container database

Creates a database container for consolidating multiple databases into a single database and enables database virtualization.

Pluggable database name:

Global Database Name is the name that identifies the database. It must be unique to distinguish the database from all other databases in a network.

The Oracle System Identifier will be the name of our database as well as serve as our **database SID**. We will need this while making connection with our database using tools such as SQLPLUS, Toad for Oracle. By default, it's set as ORCL.

If we want, we can change and give it any other name, but make sure to remember it. It's a good practice to write it down somewhere as our Database SID.

Configuration Options: Memory

Oracle Database 19c Installer - Step 9 of 17

Specify Configuration Options

19c ORACLE Database

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Memory Character sets Sample schemas

Enabling Automatic Memory Management allows the database to distribute memory automatically between the system global area (SGA) and the program global area (PGA), based on user-specified overall database memory target size. If automatic memory management is not enabled, then the SGA and PGA must be sized manually.

☒ Enable Automatic Memory Management

Allocate memory: 256 6520 16299 6,520 40 %

SGA target:	AUTO	MB
PGA aggregate target:	AUTO	MB
Target database memory:	6520	MB

Oracle Database 19c Installer - Step 9 of 17

Specify Configuration Options

19c ORACLE Database

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Memory Character sets Sample schemas

Enabling Automatic Memory Management allows the database to distribute memory automatically between the system global area (SGA) and the program global area (PGA), based on user-specified overall database memory target size. If automatic memory management is not enabled, then the SGA and PGA must be sized manually.

☐ Enable Automatic Memory Management

Allocate memory: 256 6520 16299 6,520 40 %

SGA target:	4890	MB
PGA aggregate target:	1630	MB
Target database memory:	6520	MB

Oracle Database 19c Installer - Step 9 of 17

Specify Configuration Options

19c ORACLE Database

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Memory Character sets Sample schemas

The database character set determines how character data is stored in the database.

☐ Use Unicode (AL32UTF8)

Setting character set to Unicode (AL32UTF8) enables you to store multiple language groups.

☐ Use OS character set (WE8MSWIN1252)

Character set is based on the language setting of this operating system: WE8MSWIN1252.

☒ Choose from the following list of character sets

Select Database character set:

AR8MSWIN1256 - MS Windows Code Page 1256 8-Bit Latin/Arabic

Oracle Database 19c Installer - Step 9 of 17

Specify Configuration Options

19c ORACLE Database

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Memory Character sets Sample schemas

You can choose to install sample schema in the starter database. If you choose this option, Human Resources schema will be installed in the starter database, in the SYSAUX tablespace.

☒ Install sample schemas in the database

Oracle Database 19c Installer - Step 10 of 17

Specify Database Storage Options

19c ORACLE Database

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

File system

Use a file system for database storage. For optimal database organization and performance, Oracle recommends that you install data files and the Oracle database software on different disks.

Specify database file location: D:\app\Jama_loss\oradata Browse...

Oracle Automatic Storage Management

Oracle Automatic Storage Management (Oracle ASM) simplifies database storage administration, and places database files for optimal I/O performance. Select this option if you intend to use Oracle ASM.

Oracle Database 19c Installer - Step 11 of 17

Specify Management Options

19c ORACLE Database

Oracle Database 12c is managed by Oracle Database Express by default. You can use Oracle Enterprise Manager 12c Cloud Control to manage each Oracle Database 19c centrally. Specify the details of the Cloud Control configuration to manage your database.

☐ Register with Enterprise Manager (EM) Cloud Control

OMS Host:

OMS Port:

EM Admin User Name:

EM Admin Password:

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Oracle Database 19c Installer - Step 12 of 17

Specify Recovery Options

19c ORACLE Database

Enable or disable recovery for your database. If you choose to enable recovery, then the location specified will be used as recovery area storage.

☒ Enable Recovery

☒ File system

Recovery area location:

☐ Oracle Automatic Storage Management

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Oracle Database 19c Installer - Step 13 of 17

Specify Schema Passwords

19c ORACLE Database

The starter database contains pre-loaded schemas, most of which have passwords that are expired and locked at the end of installation. After installation is complete, you must unlock and set new passwords for those accounts you want to use. Schemas used for database management and postinstallation functions are left unlocked, and passwords for these accounts will not expire. Specify the passwords for these accounts.

☒ Use different passwords for these accounts

	Password	Confirm password
SYS		
SYSTEM		
PDBADMIN		

☐ Use the same password for all accounts

Password: Confirm password:

Configuration Option

System Class

Install Type

Database Edition

Oracle Home User

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Database Storage

Management Options

Recovery Options

Schema Passwords

Minimum Password Requirements

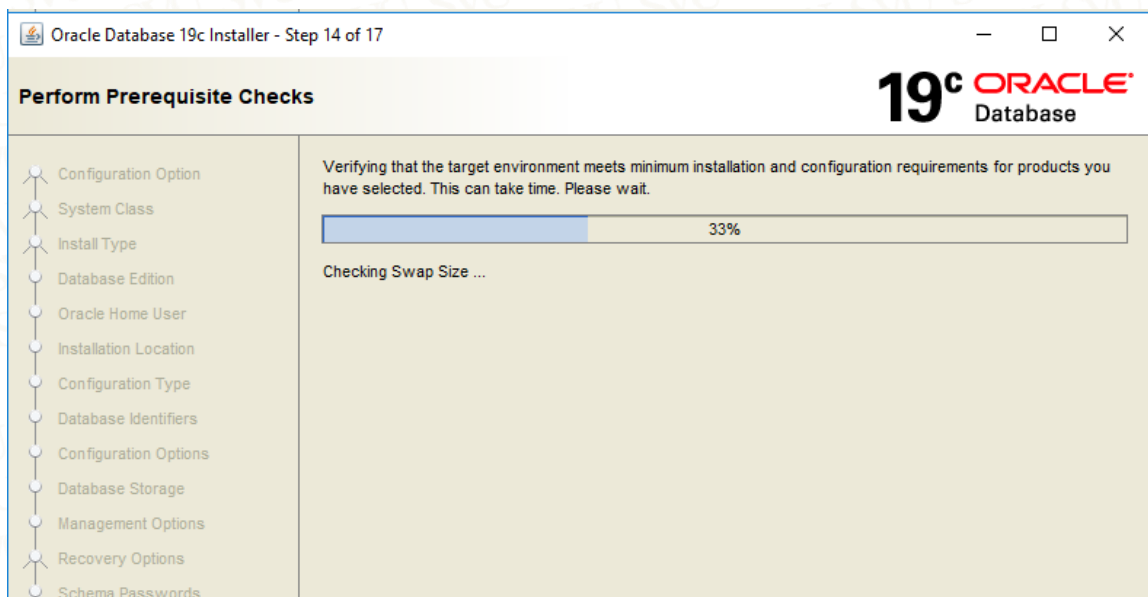
We must provide usernames and passwords that comply with the following requirements:

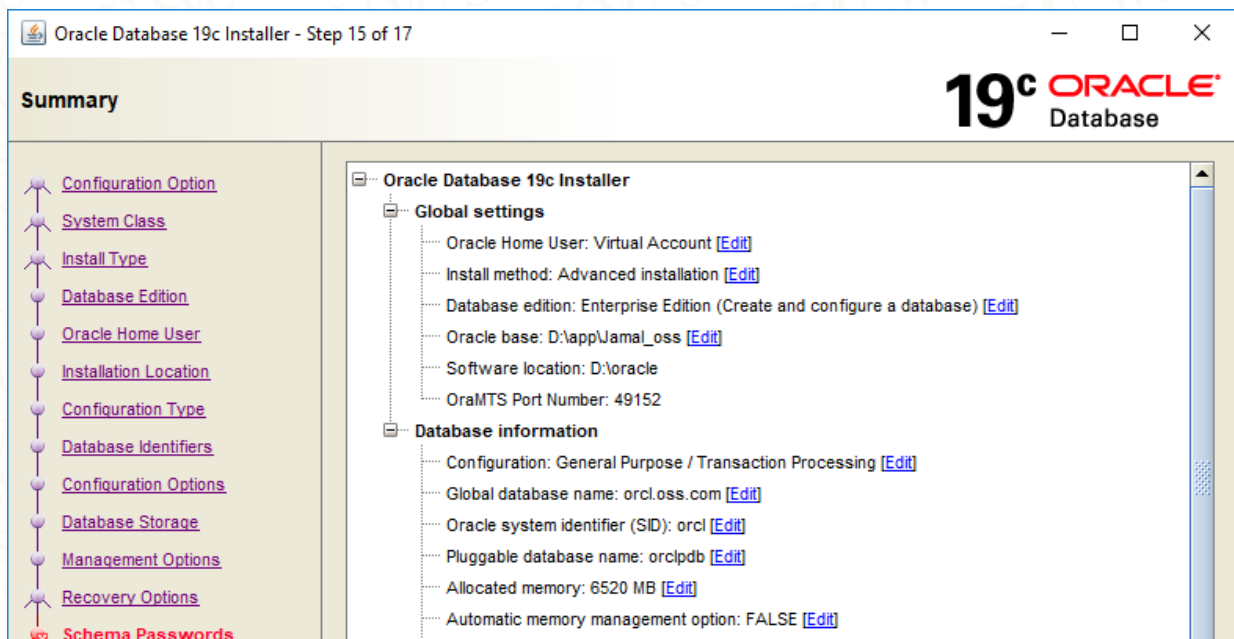
- Password cannot exceed 30 characters
- No empty password fields
- Username cannot be the same character string as a password
- The SYS account password cannot be change_on_install (case-insensitive)
- Password Recommendations

Oracle recommends the following guidelines when prompted for a password.

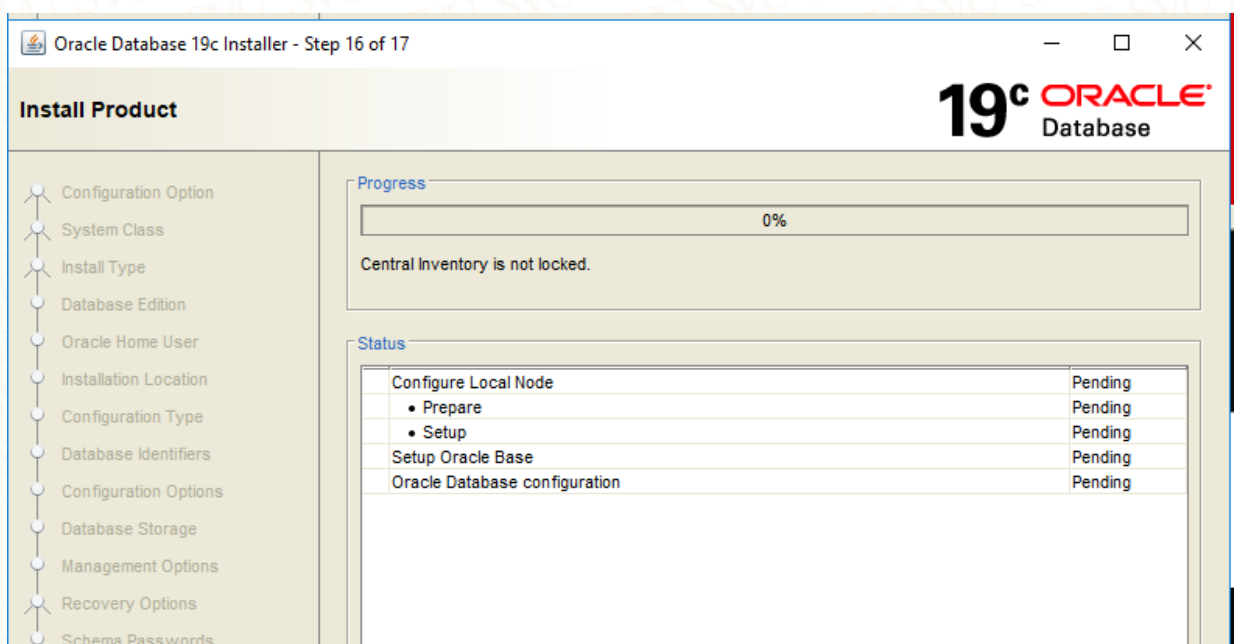
- Contains at least one lowercase letter
- Contains at least one uppercase letter
- Contains at least one digit
- Is at least 8 characters in length
- Uses the database character set, which can include the underscore (_), dollar (\$), and pound sign (#) characters.
- Should not be an actual word

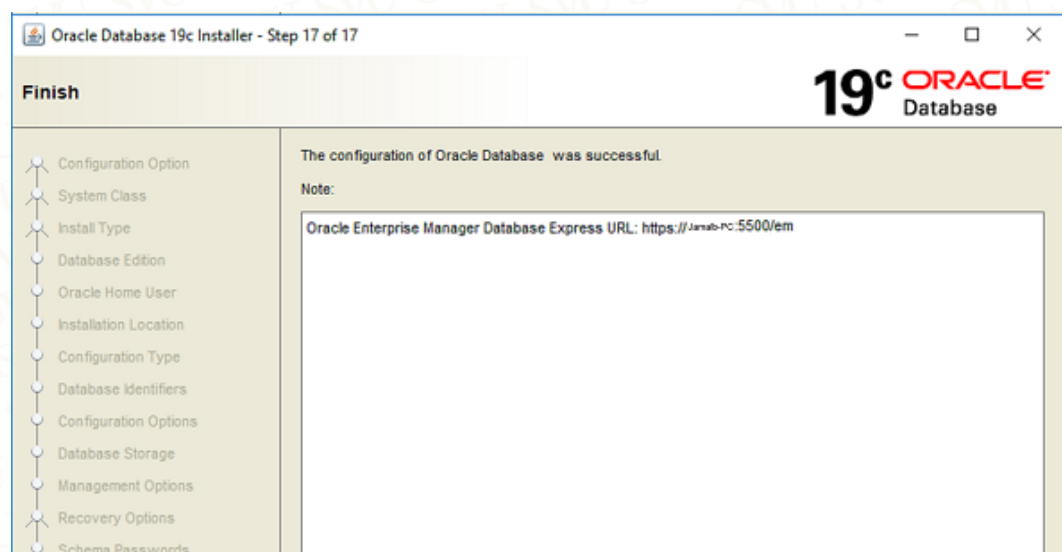
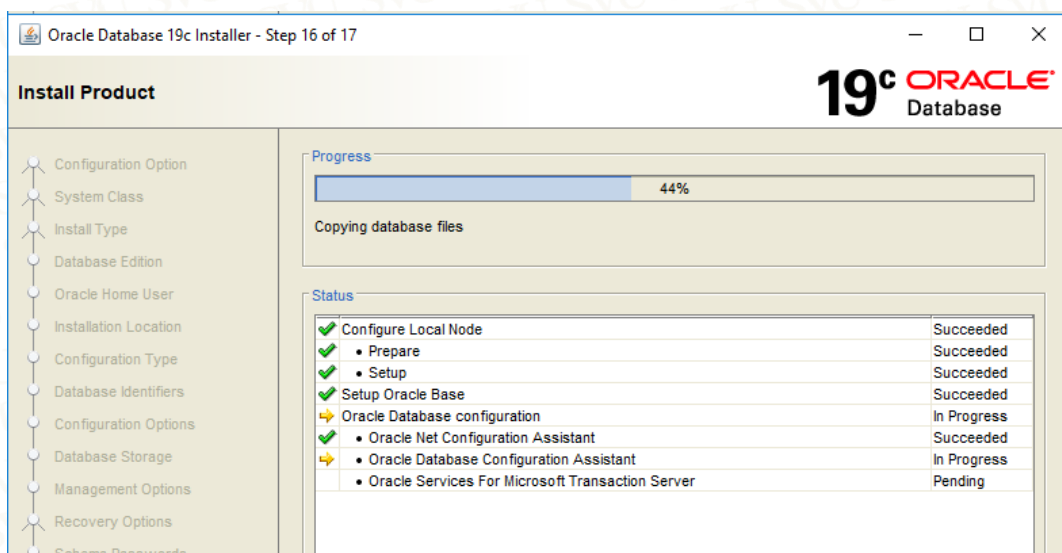
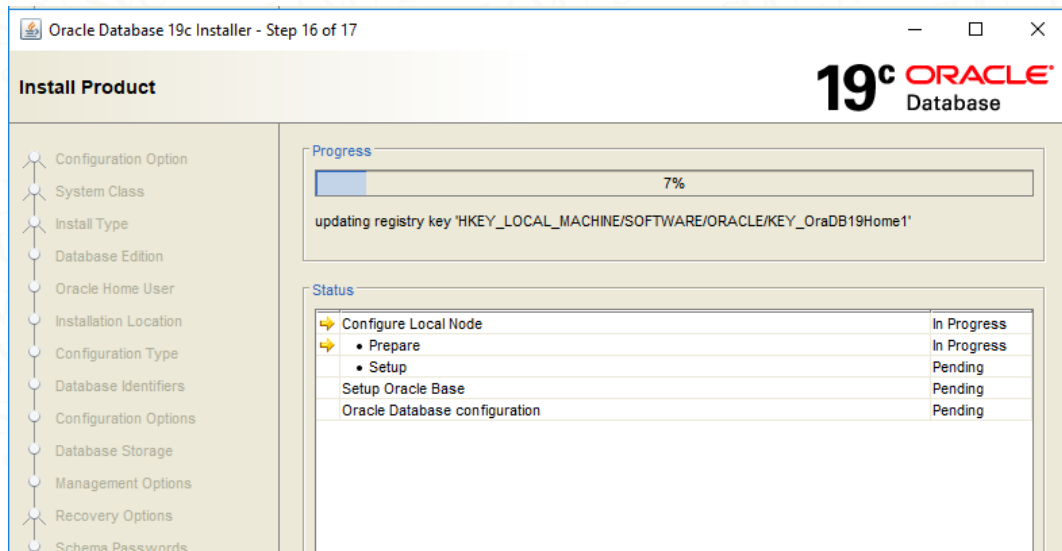
Oracle Universal Installer (OUI) will check the prerequisites, such as hardware compatibility:





The installation of Oracle Database 19c will start: this installation will take some time depending on our hardware.





The installation is done.

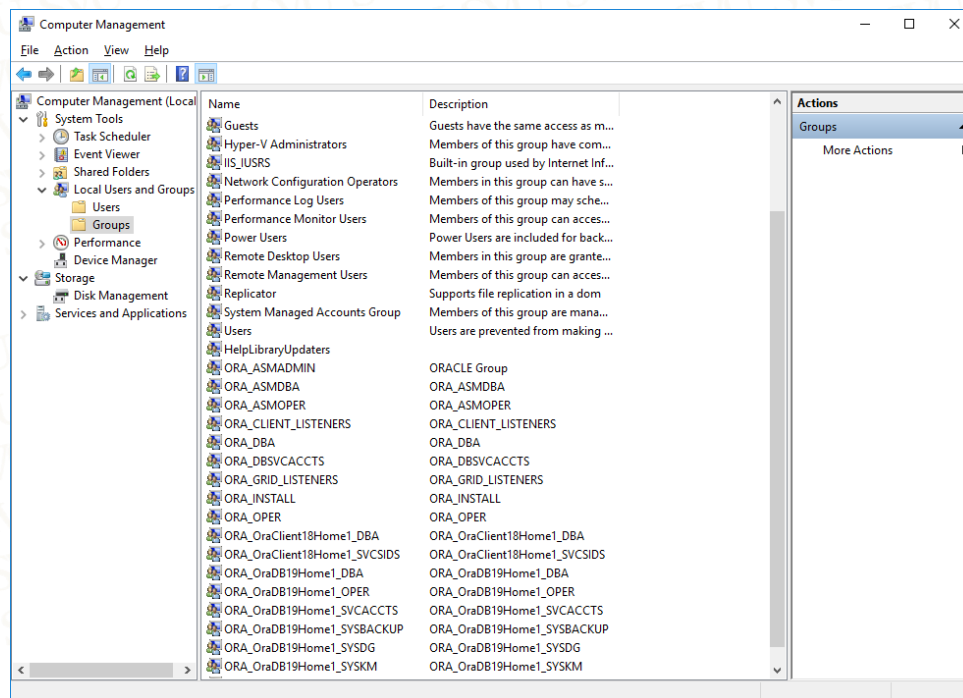
The last thing that we need to do is to copy and save the link:

<https://localhost:5500/em>

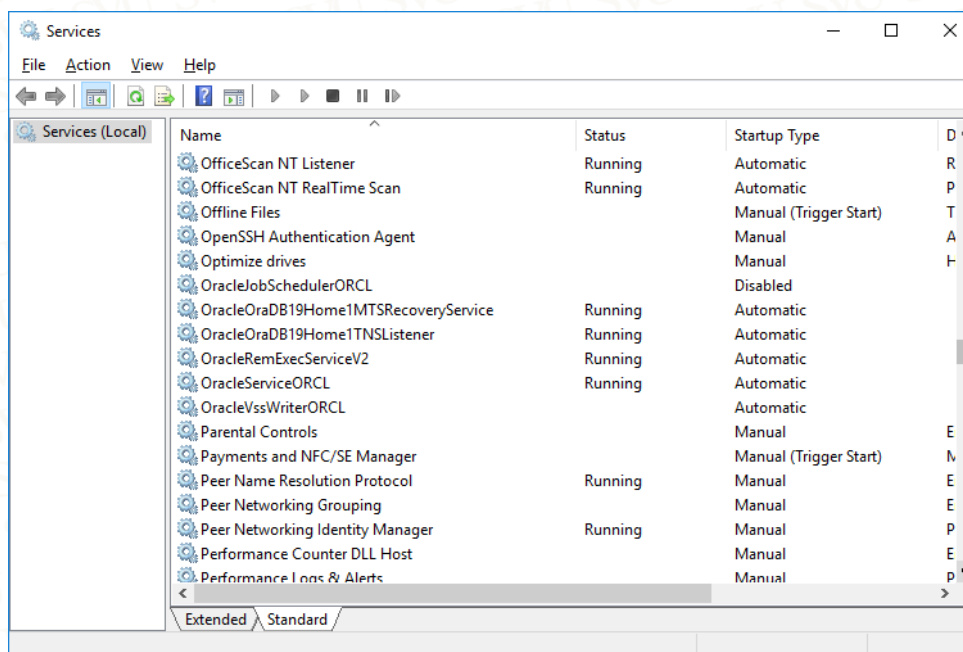
This is the link for Oracle Database 19c Enterprise Manager

Some Operating System Groups Created During Oracle Database Installation:

Operating System Group	Related System Privilege	Description
ORA_DBA	SYSDBA system privileges for all Oracle Database installations on the server	A special OSDBA group for the Windows operating system. Members of this group are granted SYSDBA system privileges for all Oracle Databases installed on the server.
ORA_OPER	SYSOPER system privileges for all Oracle Databases installed on the server	A special OSOPER group for the Windows operating system. Members of this group are granted SYSOPER system privileges all Oracle Databases installed on the server. This group does not have any members after installation, but you can manually add users to this group after the installation completes.



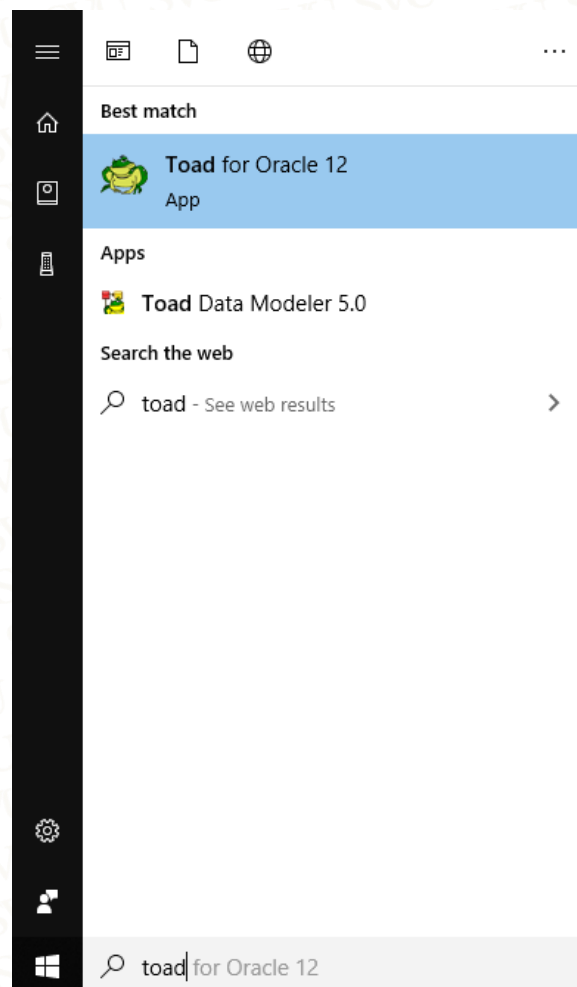
Some Operating System Services Created During Oracle Database Installation:



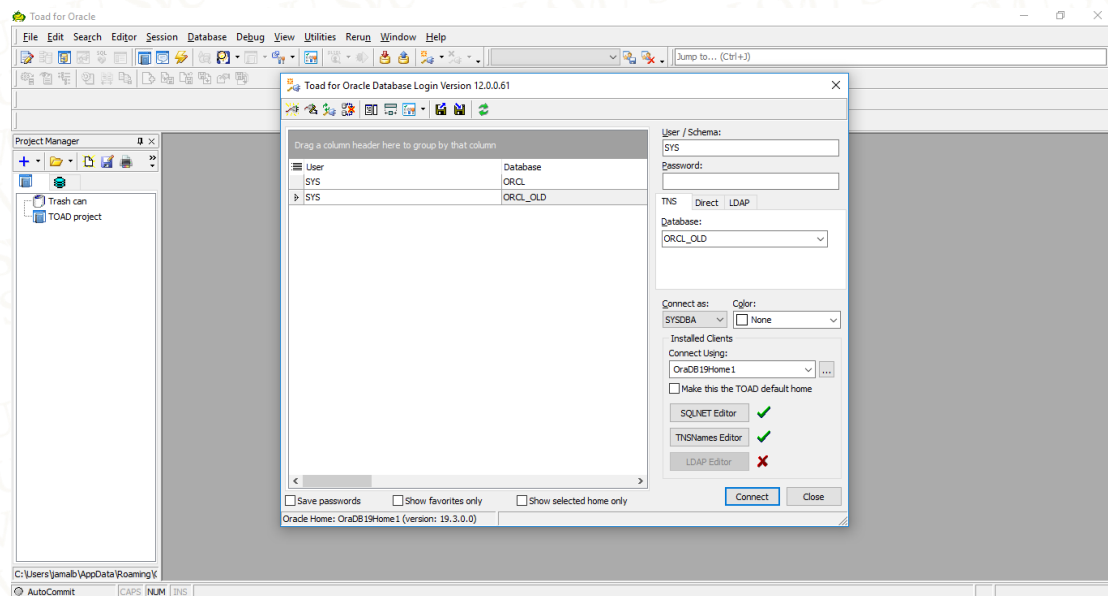
5. User's Guide for: Toad for Oracle:

We will use the tool: Toad for Oracle – 64 bit any version after Toad 12 as a principal tool to apply the database administration tasks. This is a very simple tool and has a lot of support hints and help to be satisfied during administration of your data base. In the following figures, there is a description to use Toad for Oracle step by step. The version Toad for Oracle 12 is accessible by all of the students in the first page of this subject on LMS on the SUV web site.

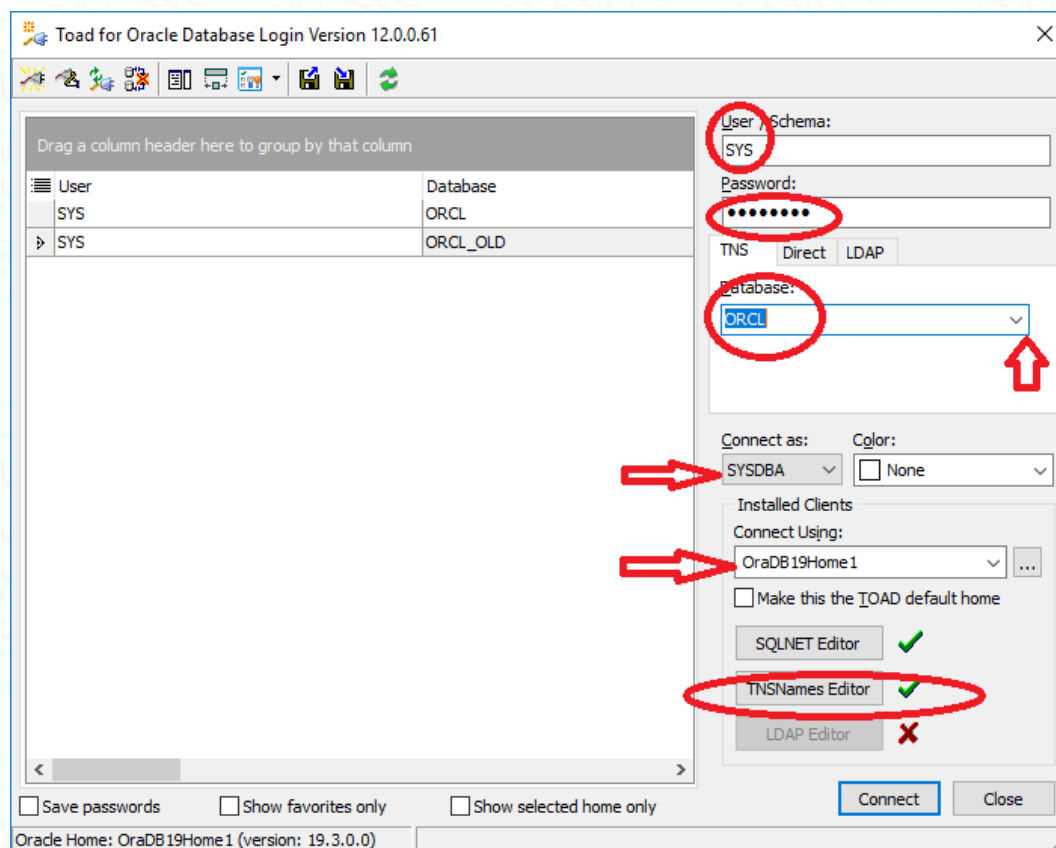
– After installation of Oracle 19c – 64 bits described here above; and installation of Toad for Oracle, you will run it from start button in the windows 10 – 64bits as the following:



You will prompt the following page:

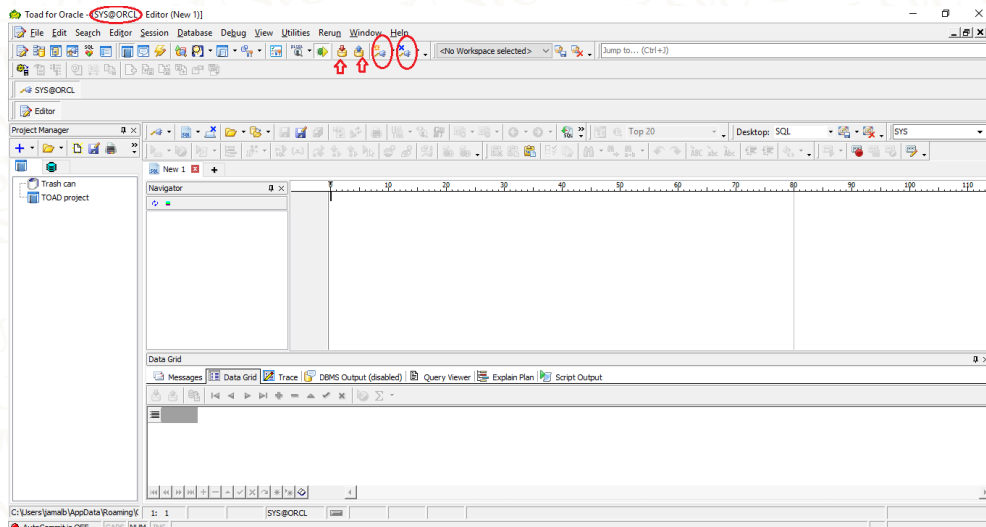


So enter your user name “SYS” and its related your Password, and choose the database from the mentioned drop list beside the field of “ Database”. And as the user is “SYS” you have to select the value for the field “Connect as” “SYSDBA”.

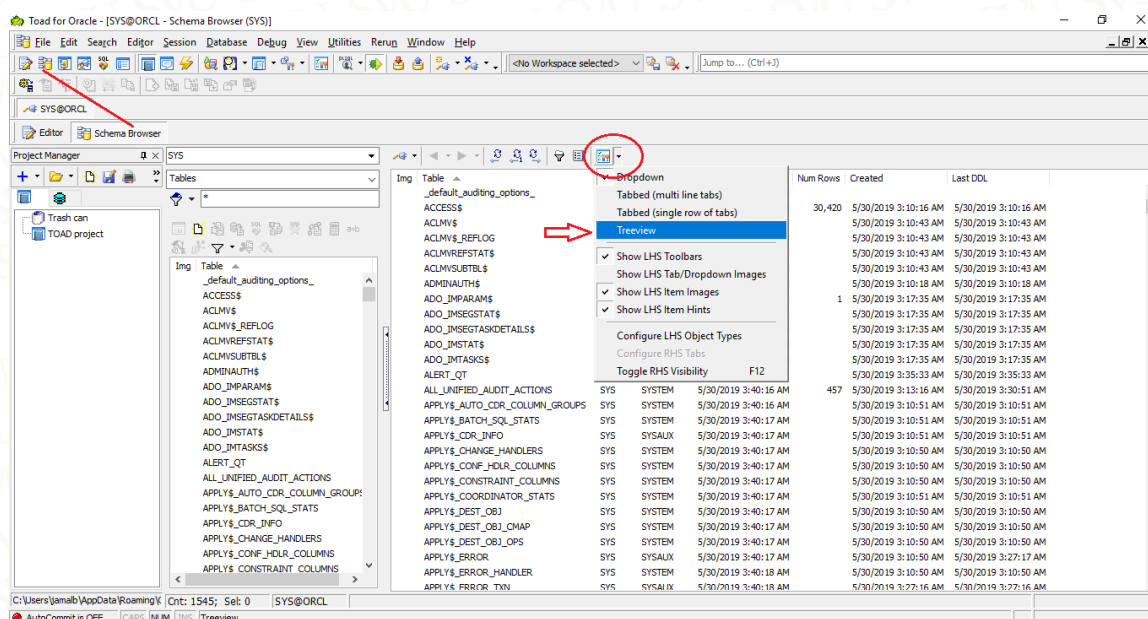


You can change the tnsnames.ora by pressing the button “ TNSNames Editor” and apply the changes or just review it if you want as a fast way to access this file.

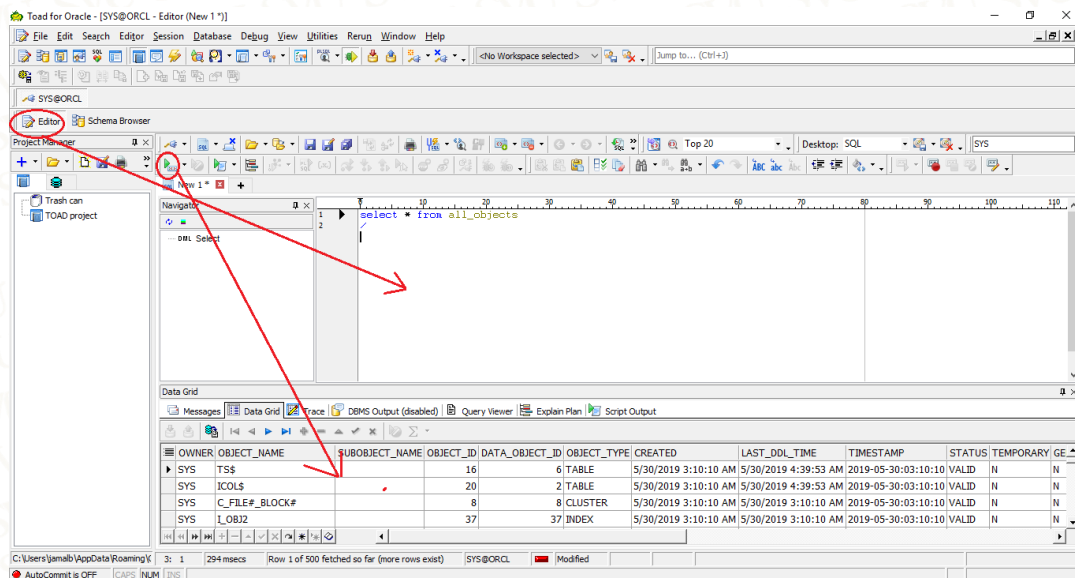
In the following step you can see the Toad for Oracle is ready to receive the various commands from your side. Notice that the current connection credential (SYS@ORCL) is appeared and the buttons of commit, rollback, new connections and disconnect the current connection are seen in the following image clearly:



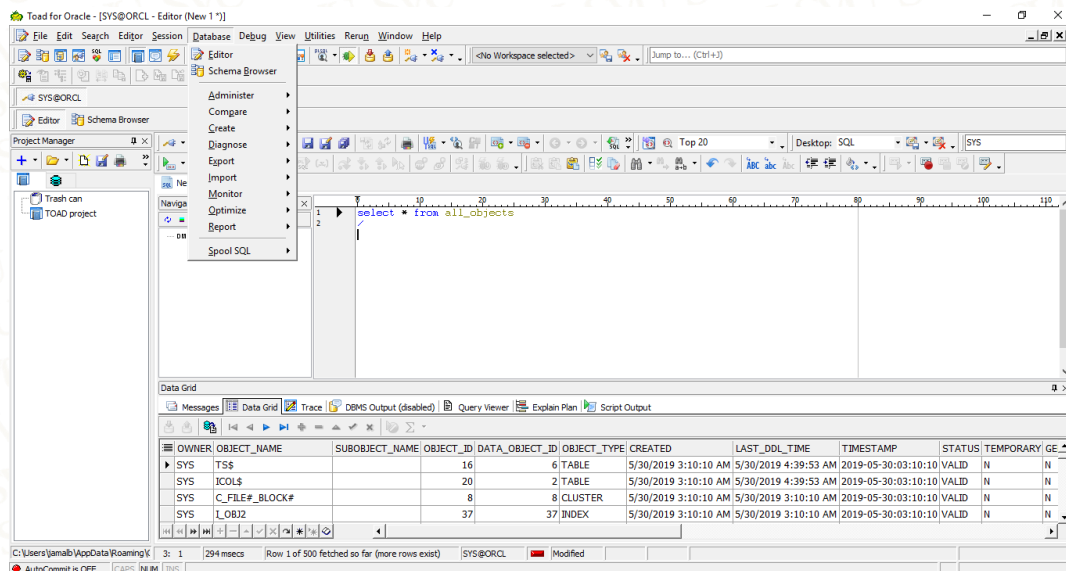
There is a very useful tool in toad for Oracle; which is “Schema Browser” which let you access all the objects of the current user/Schema and other shcema objects in a tree view to be very simple to access any object from any schema in one view:



Now, return to “Editor” page which is used to write the SQL Statement or PL/SQL statements, procedures, functions, ...etc. you can open multiple “Editor” pages as you want and save each one as a separate folder on your operating system file system:



For administrator and monitor the connected database you can find the below menu items:



In this quick user's Guide, you can continue explore the huge and simple capabilities of “Toad for Oracle” by yourself.

6. Quiz:

1. Which of these tools is not usually installed with the Oracle Universal Installer?
(Choose the best answer).
 - a) The Oracle Universal Installer itself
 - b) SQL*Plus
 - c) SQL Developer
 - d) Oracle Enterprise Express Manager
2. What statement best describes the relationship between the Oracle Base and the Oracle Home? (Choose the best answer).
 - a) The Oracle Base exists inside the Oracle Home.
 - b) The Oracle Base can contain Oracle Homes for different products.
 - c) One Oracle Base is required for each product, but versions of the product can exist in their own Oracle Homes within their Oracle Base.
 - d) The Oracle Base is created when you run the orainstRoot.sh script, and contains a pointer to the Oracle Home.
3. What does Optimal Flexible Architecture (OFA) describe? (Choose the best answer).
 - a) A directory structure
 - b) Distributed database systems
 - c) Multitier processing architecture
 - d) OFA encompasses all the above
4. What is the statement that is not a DBA task?
 - a) Evaluating the database server hardware
 - b) Installing the Oracle software
 - c) Shutdown and boot the OS
 - d) Planning the database and security strategy

5. Monitoring database performance is not a DBA task?
 - a) True
 - b) False
6. ORACLE_HOME directories for read-only directories?
 - a) True
 - b) False
7. ORACLE_HOME directories for the writable files in the ORACLE_BASE?
 - a) True
 - b) False
8. Which one of the following environment variables is not related to Oracle database server?
 - a) ORACLE_HOME
 - b) ORACLE_BASE
 - c) ORACLE_SID
 - d) W2kVersion
9. Members of ORA_DBA group are granted SYSDBA system privileges for all Oracle Databases installed on the server?
 - a) True
 - b) False
10. Which is the group that its members are granted SYSDBA system privileges for all Oracle Databases installed on the server?
 - a) ORA_OPER
 - b) ORA_INSTALL
 - c) ORA_ASMOPER
 - d) ORA_DBA

11. Members of ORA_DBA group are granted CREATE TABLE system privilege only for Oracle Database?
- a) True
 - b) False
12. Oracle Database 19c is only available for?
- a) 32 bit Windows systems
 - b) 64 bit Windows systems
 - c) Windows XP
 - d) Windows Millinium
13. Oracle Database 19c is only available for 64 bit Windows systems?
- a) True
 - b) False
14. What is the sufficient Physical memory (RAM) to run Oracle alone:
- a) 1 GB
 - b) 4 GB
 - c) 2GB
 - d) 8 GB
15. The configuration of Virtual memory (swap) size has relation with the size of:
- a) Physical memory
 - b) Hard Drive
 - c) CD Drive
 - d) PGA

16. Oracle Database 19c is only available for Windows 10 Professional, Enterprise and Education Editions?
- a) True
 - b) False
17. Oracle Database 19c is only available for Windows 10 Home Edition?
- a) True
 - b) False
18. We can't install Oracle Database 19c if we are using:
- a) Windows 10 Professional
 - b) Enterprise Edition
 - c) Education Edition
 - d) Windows 10 Home Edition
19. The link <https://localhost:5500/em> is for:
- a) Oracle Database Configuration Assistance
 - b) Oracle Database 19c Enterprise Manager
 - c) Oracle Network Configuration Assistance
20. Oracle 19c is not available for any Linux or Unix operating system?
- a) True
 - b) False

Quiz answers – chapter 2	
1	c
2	b
3	a
4	c
5	b
6	a
7	b
8	d
9	a
10	d
11	b
12	b
13	a
14	c
15	a
16	a
17	b
18	d
19	b
20	b